

Oscillatory Dynamics In Cell Signaling

Samerah Al Mosa & Elisha Erzoah

George Mason University

November 21, 2025



Outline

1 Introduction

2 Models Exhibiting Oscillations

- A model with Delay Differential Equations
- Extending the Model with Ordinary Differential Equations

3 Conclusion

Introduction

- ▶ **In neurogenesis, Lateral Inhibition plays a critical role in cell fate determination.**

This lateral inhibition mechanism creates **oscillations** in cellular activities.

Introduction

- ▶ **In neurogenesis, Lateral Inhibition plays a critical role in cell fate determination.**

This lateral inhibition mechanism creates **oscillations** in cellular activities.

- ▶ **Mathematical models** allow us to explore:
 - How these oscillations arise from feedback loops
 - What sustains these oscillations over time
 - How nonlinear interactions shape their dynamics

- ▶ **In neurogenesis, Lateral Inhibition plays a critical role in cell fate determination.**

This lateral inhibition mechanism creates **oscillations** in cellular activities.

- ▶ **Mathematical models** allow us to explore:
 - How these oscillations arise from feedback loops
 - What sustains these oscillations over time
 - How nonlinear interactions shape their dynamics
- ▶ **In this talk**, we examine the mathematical principles behind oscillations in cell signaling networks.

Two-Cell Mutual Repression

Considering the model below;

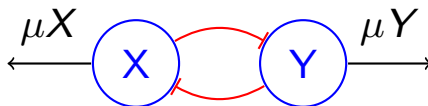


Figure: A schematic representation of two-cell mutual repression. (—→) represents decay whilst (—|) represents down-regulation or repression. X and Y are the Cells.

► Modeling this cell signaling motif with Delay Differential Equations(DDEs) exhibits oscillations (**Glass et al. 2021**).

Model Description

The delay differential equations(DDEs) representing this model is given by;

$$\frac{dX}{dt} = -\mu X + (1 + b)f(Y(t - \tau)) \quad (1)$$

$$\frac{dY}{dt} = -\mu Y + f(X(t - \tau)), \quad (2)$$

where:

- $X(t)$ and $Y(t)$: Activity levels of cells X and Y .
- μ : Decay rate (positive constant).
- b : *bias parameter*.
- τ : Time delay .
- $f(\cdot)$: Monotone decreasing repression function.

Considering a Case without Time Delay ($\tau = 0$)

When we set ($\tau = 0$) , the model becomes:

$$\frac{dX}{dt} = -\mu X + (1 + b) f(Y), \quad (3)$$

$$\frac{dY}{dt} = -\mu Y + f(X), \quad (4)$$

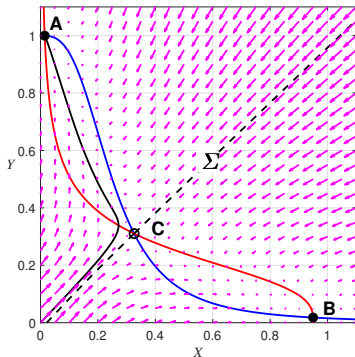
where $f(Z) = \frac{\theta^n}{\theta^n + Z^n}$ ($\theta > 0, n > 0$)

The steady states (X_* , Y_*) of the system are given by,

$$X_* = \frac{1}{\mu}(1 + b) f(Y_*), \quad (5)$$

$$Y_* = \frac{1}{\mu} f(X_*). \quad (6)$$

Phase Potrait



- ▶ The parameters used are $\mu = 1$, $\theta = 0.25$, $n = 3$, and $b = -0.05$
- ▶ The red and blue lines represent the X - and Y -nullclines, respectively.
- ▶ The arrows indicate the vector field $\left(\frac{dX}{dt}, \frac{dY}{dt}\right)$, showing the local directions of trajectories in the (X, Y) .
- ▶ longer arrows correspond to faster trajectories approaching a steady state..

Slopes at steady states

The slope of the nullclines at the steady points are given by,

The X-nullcline:

$$X = \frac{1}{\mu}(1+b)f(Y).$$

where

$$\frac{dX}{dY} = \frac{1}{\mu}(1+b)f'(Y)$$

Since $f'(Y) < 0$, we define:

$$\phi_2 = \left. \frac{df}{dY} \right|_{Y_*} < 0.$$

To find the slope on the phase plane (with Y vs X), we invert

$$\frac{dY}{dX} = \frac{\mu}{(1+b)\phi_2}.$$

Slopes at steady states

The Y-nullcline:

$$Y = \frac{1}{\mu} f(X).$$

where

$$\frac{dY}{dX} = \frac{1}{\mu} f'(X).$$

We define:

$$\phi_1 = \left. \frac{df}{dX} \right|_{X_*} < 0.$$

The slope of the Y-nullcline is:

$$\frac{dY}{dX} = \frac{1}{\mu} \phi_1.$$

Slopes at steady states

We represent the slope of each nullcline as

$$\gamma_1 = \frac{\mu}{(1+b)\phi_2}, \quad \gamma_2 = \frac{1}{\mu}\phi_1.$$

- At steady states A and B , we have $\gamma_1 < \gamma_2$, which gives

$$\frac{\mu}{(1+b)\phi_2} < \frac{1}{\mu}\phi_1 \quad \Rightarrow \quad \mu^2 > (1+b)\phi_1\phi_2.$$

- At the steady state C , $\gamma_1 > \gamma_2$, and we have

$$\frac{\mu}{(1+b)\phi_2} > \frac{1}{\mu}\phi_1 \quad \Rightarrow \quad \mu^2 < (1+b)\phi_1\phi_2.$$

Linearized system

We take small perturbations from the steady state:

$$x = X - X_*, \quad y = Y - Y_*,$$

and linearize the original ODEs at the steady state we have:

$$\frac{dx}{dt} = -\mu x + (1 + b) \phi_2 y$$

$$\frac{dy}{dt} = -\mu y + \phi_1 x$$

The Jacobian matrix is,

$$J = \begin{bmatrix} -\mu & (1 + b) \phi_2 \\ \phi_1 & -\mu \end{bmatrix},$$

The characteristic equation of J is:

$$(\mu + \lambda)^2 = (1 + b) \phi_1 \phi_2.$$

Linearized system

The eigenvalues are

$$\lambda_1 = -\mu + \chi \quad (7)$$

$$\lambda_2 = -\mu - \chi \quad (8)$$

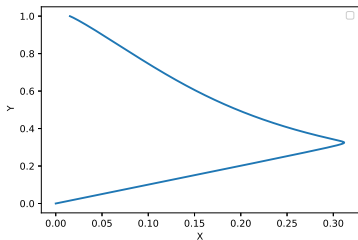
Where $\chi = \sqrt{(1+b)\phi_1\phi_2}$; (w.l.o.g $\chi > 0$). where χ represents the sensitivity of the system.

We have two possible cases:

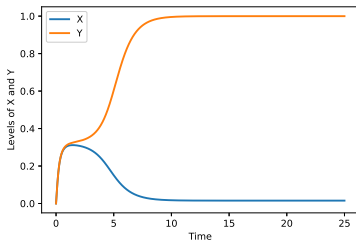
- If $\chi < \mu$: both eigenvalues are real and negative, so the steady states A and B are **stable** and we do not have switching.
- If $\chi > \mu$: One eigenvalue becomes positive(i.e λ_2), so C is **unstable** (a saddle point). When the system reaches C , it must leave and move to either A or B , depending on the initial condition.

Behaviour of the System ($\tau = 0$)

Phase-Plane and Time Dynamics Showing Switching Behaviour



(a)



(b)

(a) The trajectory starting from $(0,0)$ approaches the unstable steady state and then switches to the stable state A.

(b) Time courses of X and Y showing a rapid transition followed by convergence to high Y and low X .

Considering a Case with Time Delay ($\tau > 0$)

Obtaining the Solutions of the DDEs

- The delay differential equations(DDEs) representing this model is given by;

$$\frac{dX}{dt} = -\mu X + (1 + b)f(Y(t - \tau)) \quad (9)$$

$$\frac{dY}{dt} = -\mu Y + f(X(t - \tau)), \quad (10)$$

Considering a Case with Time Delay ($\tau > 0$)

Obtaining the Solutions of the DDEs

- The delay differential equations(DDEs) representing this model is given by;

$$\frac{dX}{dt} = -\mu X + (1 + b)f(Y(t - \tau)) \quad (9)$$

$$\frac{dY}{dt} = -\mu Y + f(X(t - \tau)), \quad (10)$$

- Linearising the DDEs around a steady state of the system in (9) and (10), we have;

$$\frac{dx}{dt} = -\mu x + (1 + b)\phi_2 y(t - \tau) \quad (11a)$$

$$\frac{dy}{dt} = -\mu y + \phi_1 x(t - \tau) \quad (11b)$$

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We will consider the second derivative of x to determine solutions.

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} + (1+b)\phi_2 \frac{dy}{dt}(t-\tau) \quad (12)$$

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We will consider the second derivative of x to determine solutions.

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} + (1+b)\phi_2 \frac{dy}{dt}(t-\tau) \quad (12)$$

Taking time backwards in eqn $\frac{dy}{dt} = -\mu y + \phi_1 x(t-\tau)$ from (11b), we have

$$\frac{dy(t-\tau)}{dt} = -\mu y(t-\tau) + \phi_1 x(t-2\tau). \quad (13)$$

Considering a Case with Time Delay ($\tau > 0$)

- We will consider the second derivative of x to determine solutions.

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} + (1+b)\phi_2 \frac{dy}{dt}(t-\tau) \quad (12)$$

Taking time backwards in eqn $\frac{dy}{dt} = -\mu y + \phi_1 x(t-\tau)$ from (11b), we have

$$\frac{dy(t-\tau)}{dt} = -\mu y(t-\tau) + \phi_1 x(t-2\tau). \quad (13)$$

Substituting (13) into (12)

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} - \mu(1+b)\phi_2 y(t-\tau) + (1+b)\phi_1\phi_2 x(t-2\tau) \quad (14)$$

Considering a Case with Time Delay ($\tau > 0$)

- We will consider the second derivative of x to determine solutions.

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} + (1+b)\phi_2 \frac{dy}{dt}(t-\tau) \quad (12)$$

Taking time backwards in eqn $\frac{dy}{dt} = -\mu y + \phi_1 x(t-\tau)$ from (11b), we have

$$\frac{dy(t-\tau)}{dt} = -\mu y(t-\tau) + \phi_1 x(t-2\tau). \quad (13)$$

Substituting (13) into (12)

$$\frac{d^2x}{dt^2} = -\mu \frac{dx}{dt} - \mu(1+b)\phi_2 y(t-\tau) + (1+b)\phi_1\phi_2 x(t-2\tau) \quad (14)$$

Substituting $(1+b)\phi_2 y(t-\tau) = \frac{dx}{dt} + \mu x$ from (11a) into (14) and simplifying we obtain the expression.

Considering a Case with Time Delay ($\tau > 0$)

$$\frac{d^2x}{dt^2} + 2\mu \frac{dx}{dt} + \mu^2 x = \chi^2 x(t - 2\tau) \quad (15)$$

Where $\chi = \sqrt{(1+b)\phi_1\phi_2}$; (w.l.o.g $\chi > 0$).

Considering a Case with Time Delay ($\tau > 0$)

$$\frac{d^2x}{dt^2} + 2\mu \frac{dx}{dt} + \mu^2 x = \chi^2 x(t - 2\tau) \quad (15)$$

Where $\chi = \sqrt{(1+b)\phi_1\phi_2}$; (w.l.o.g $\chi > 0$).

Now we seek the solutions of the form $x(t) = x_0 e^{(\lambda t)}$ where $\lambda \in \mathbb{C}$.

$$\begin{aligned}\lambda^2 + 2\mu\lambda + \mu^2 &= \chi^2 e^{-2\lambda\tau} \\ (\lambda + \mu)^2 &= (\chi e^{-\lambda\tau})^2 \\ \lambda + \mu &= \pm \chi e^{-\lambda\tau}\end{aligned}$$

Considering a Case with Time Delay ($\tau > 0$)

$$\frac{d^2x}{dt^2} + 2\mu \frac{dx}{dt} + \mu^2 x = \chi^2 x(t - 2\tau) \quad (15)$$

Where $\chi = \sqrt{(1+b)\phi_1\phi_2}$; (w.l.o.g $\chi > 0$).

Now we seek the solutions of the form $x(t) = x_0 e^{(\lambda t)}$ where $\lambda \in \mathbb{C}$.

$$\begin{aligned}\lambda^2 + 2\mu\lambda + \mu^2 &= \chi^2 e^{-2\lambda\tau} \\ (\lambda + \mu)^2 &= (\chi e^{-\lambda\tau})^2 \\ \lambda + \mu &= \pm \chi e^{-\lambda\tau}\end{aligned}$$

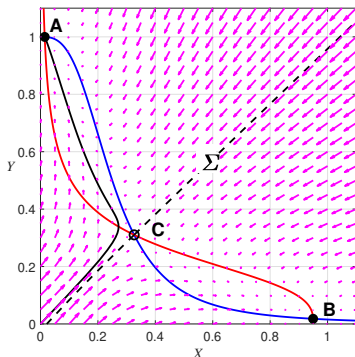
The Solutions

$$\lambda_1 + \mu = \chi e^{-\lambda_1\tau} \quad (16a)$$

$$\lambda_2 + \mu = -\chi e^{-\lambda_2\tau} \quad (16b)$$

Considering a Case with Time Delay ($\tau > 0$)

The Time Taken to Reach the Stable States (Switching Time)



- ▶ The time taken to reach State A is governed by the equation (16a), $\lambda_1 + \mu = \chi e^{-\lambda_1 \tau}$.
- ▶ Time delay increases the time to reach steady state A
- ▶ Time delay reduces the speed of the trajectory to reach steady state A.

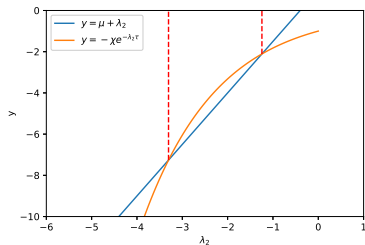
The Speeding Rate is Approximated as

$$\lambda_1 > \frac{\chi - \mu}{1 + \chi \tau}$$

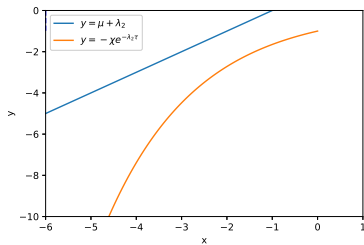
Considering a Case with Time Delay ($\tau > 0$)

The Emergence of Oscillations

The dynamics of oscillations occur on Σ and it is characterized by equation (16b), $\lambda_2 + \mu = -\chi e^{-\lambda_2 \tau}$.



(a) For smaller τ



(b) For larger τ

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We want to find a critical τ at which $\text{Re}(\lambda_2) = 0$.

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We want to find a critical τ at which $\text{Re}(\lambda_2) = 0$.
- ▶ Assuming there is τ_H , where $\text{Re}(\lambda_2) = 0$ then $\lambda_2 = i\omega_H$, where $\omega_H \in \mathbb{R}$.

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We want to find a critical τ at which $\text{Re}(\lambda_2) = 0$.
- ▶ Assuming there is τ_H , where $\text{Re}(\lambda_2) = 0$ then $\lambda_2 = i\omega_H$, where $\omega_H \in \mathbb{R}$.

Substituting $\lambda_2 = i\omega_H$ into $\lambda_2 + \mu = -\chi e^{-\lambda_2 \tau}$ from (16b) we obtain;

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We want to find a critical τ at which $\text{Re}(\lambda_2) = 0$.
- ▶ Assuming there is τ_H , where $\text{Re}(\lambda_2) = 0$ then $\lambda_2 = i\omega_H$, where $\omega_H \in \mathbb{R}$.

Substituting $\lambda_2 = i\omega_H$ into $\lambda_2 + \mu = -\chi e^{-\lambda_2 \tau}$ from (16b) we obtain;

$$i\omega_H + \mu = -\chi e^{-i\omega_H \tau_H}$$

$$i\omega_H + \mu = -\chi(\cos(\omega_H \tau_H) - i \sin(\omega_H \tau_H)) \quad (17)$$

Considering a Case with Time Delay ($\tau > 0$)

- ▶ We want to find a critical τ at which $\text{Re}(\lambda_2) = 0$.
- ▶ Assuming there is τ_H , where $\text{Re}(\lambda_2) = 0$ then $\lambda_2 = i\omega_H$, where $\omega_H \in \mathbb{R}$.

Substituting $\lambda_2 = i\omega_H$ into $\lambda_2 + \mu = -\chi e^{-\lambda_2 \tau}$ from (16b) we obtain;

$$i\omega_H + \mu = -\chi e^{-i\omega_H \tau_H}$$

$$i\omega_H + \mu = -\chi(\cos(\omega_H \tau_H) - i \sin(\omega_H \tau_H)) \quad (17)$$

Equating the real and imaginary parts, we obtain (17);

$$\omega_H = \chi \sin(\omega_H \tau_H) \quad (18a)$$

$$\mu = -\chi \cos(\omega_H \tau_H) \quad (18b)$$

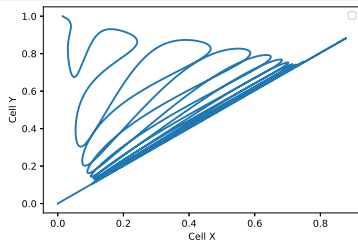
Considering a Case with Time Delay ($\tau > 0$)

The critical value (τ_H) at which the system transitions into Hopf bifurcation (purely imaginary eigen value) is;

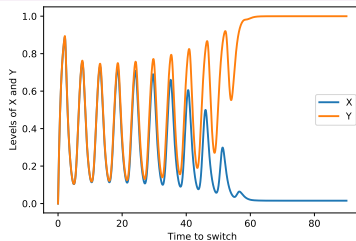
Bifurcation Value (τ_H)

$$\tau_H = \frac{1}{\omega_H} \tan^{-1} \left(\frac{-\omega_H}{\mu} \right) = \frac{1}{\omega_H} \left[\pi - \tan^{-1} \left(\frac{\omega_H}{\mu} \right) \right]$$

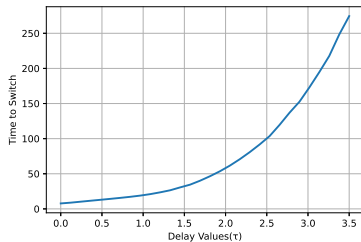
Considering a Case with Time Delay ($\tau > 0$)



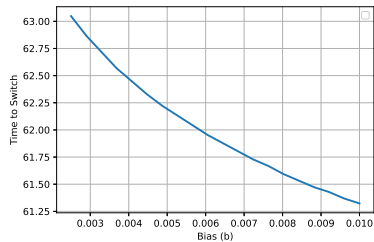
(a)



(b)



(c)



(d)

Modeling Oscillatory Dynamics with ODEs

- We extend the model from Delay Differential Equations (DDEs) to Ordinary Differential Equations (ODEs) (Gonze & Ruo 2023).

Modeling Oscillatory Dynamics with ODEs

- We extend the model from Delay Differential Equations (DDEs) to Ordinary Differential Equations (ODEs) (Gonze & Ruo 2023).
- **What is the rational?**

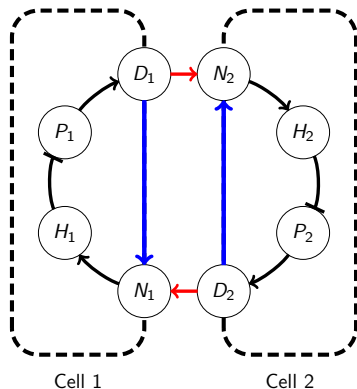
Modeling Oscillatory Dynamics with ODEs

- We extend the model from Delay Differential Equations (DDEs) to Ordinary Differential Equations (ODEs) (Gonze & Ruo 2023).
- **What is the rational?**
- Biologically, cellular repression isn't a single event but a kinetic cascade.

Modeling Oscillatory Dynamics with ODEs

- We extend the model from Delay Differential Equations (DDEs) to Ordinary Differential Equations (ODEs) (Gonze & Ruo 2023).
- **What is the rational?**
- Biologically, cellular repression isn't a single event but a kinetic cascade.
- Our DDE model summarized this entire cascade as a single pause. By extending the model to a system of ODEs, we "unpack" the delay into its constituent parts.

Modeling Oscillatory Dynamics with ODEs



- Each node in the network represents a specific class of protein, indicated by the corresponding letters. D: **Delta Ligand**, N: **Notch Receptor**, H: **Hes Proteins**, P: **Proneural Proteins**.
- The weight of these interactions in this scheme are given as; Cis interaction assumes weight p whilst trans-interaction has weight $(1 - p)$.

Modeling Oscillatory Dynamics with ODEs

- ▶ The regulation mechanism of this scheme is presented by the following system of ordinary differential equations.

Modeling Oscillatory Dynamics with ODEs

- ▶ The regulation mechanism of this scheme is presented by the following system of ordinary differential equations.

$$\frac{dN_1}{dt} = -\mu N_1 + (1 + b)f(pD_1 + (1 - p)D_2) \qquad \frac{dH_2}{dt} = -\mu H_2 + f(N_2)$$

$$\frac{dH_1}{dt} = -\mu H_1 + f(N_1) \qquad \frac{dP_2}{dt} = -\mu P_2 + g(H_2)$$

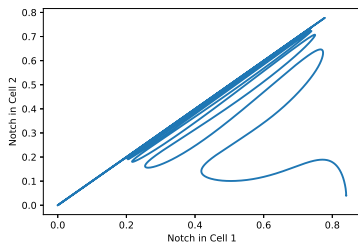
$$\frac{dP_1}{dt} = -\mu P_1 + g(H_1) \qquad \frac{dD_2}{dt} = -\mu D_2 + f(P_2)$$

$$\frac{dD_1}{dt} = -\mu D_1 + f(P_1)$$

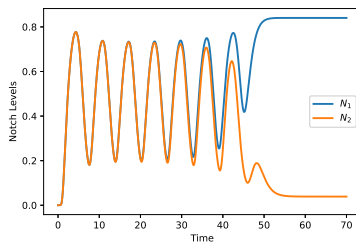
$$\frac{dN_2}{dt} = -\mu N_2 + f(pD_2 + (1 - p)D_1)$$

where $f(Z) = g(Z) = \frac{\theta^n}{\theta^n + Z^n}$

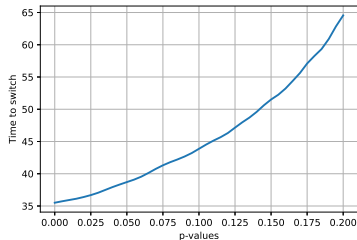
Modeling Oscillatory Dynamics with ODEs



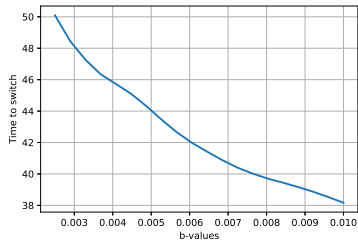
(a)



(b)



(c)



(d)

Conclusion

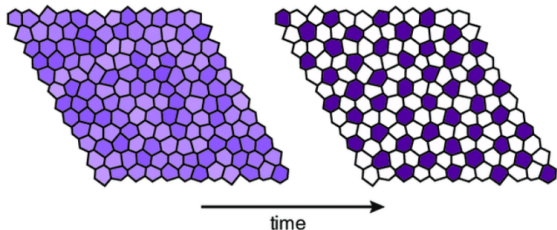
- ▶ Oscillations via lateral inhibition create enough time for the nervous system of humans to fully develop for proper functioning.

Conclusion

- ▶ Oscillations via lateral inhibition create enough time for the nervous system of humans to fully develop for proper functioning.
- ▶ Oscillations via lateral inhibition give rise to pattern formation (Collier et al., 1996).

Local cell-cell communication → Global patterns

Example of such beautiful patterns is the Zebrafish stripes



References

- Gonze, D., & Ruoff, P. (2023). The Goodwin oscillator and its legacy. *Acta Biotheoretica*, **69**, 857–874.
- Glass, D. S., Jin, X., & Riedel-Kruse, I. H. (2021). Nonlinear delay differential equations and their application to modeling biological network motifs. *Nature Communications*, **12**(1), 1788.
- Collier, J. R., Monk, N. A., Maini, P. K., & Lewis, J. H. (1996). Pattern formation by lateral inhibition with feedback: a mathematical model of delta-notch intercellular signalling. *Journal of Theoretical Biology*, **183**(4), 429–446.

Thank you for your Attention

QUESTIONS??